Critique depth — a documented near-null

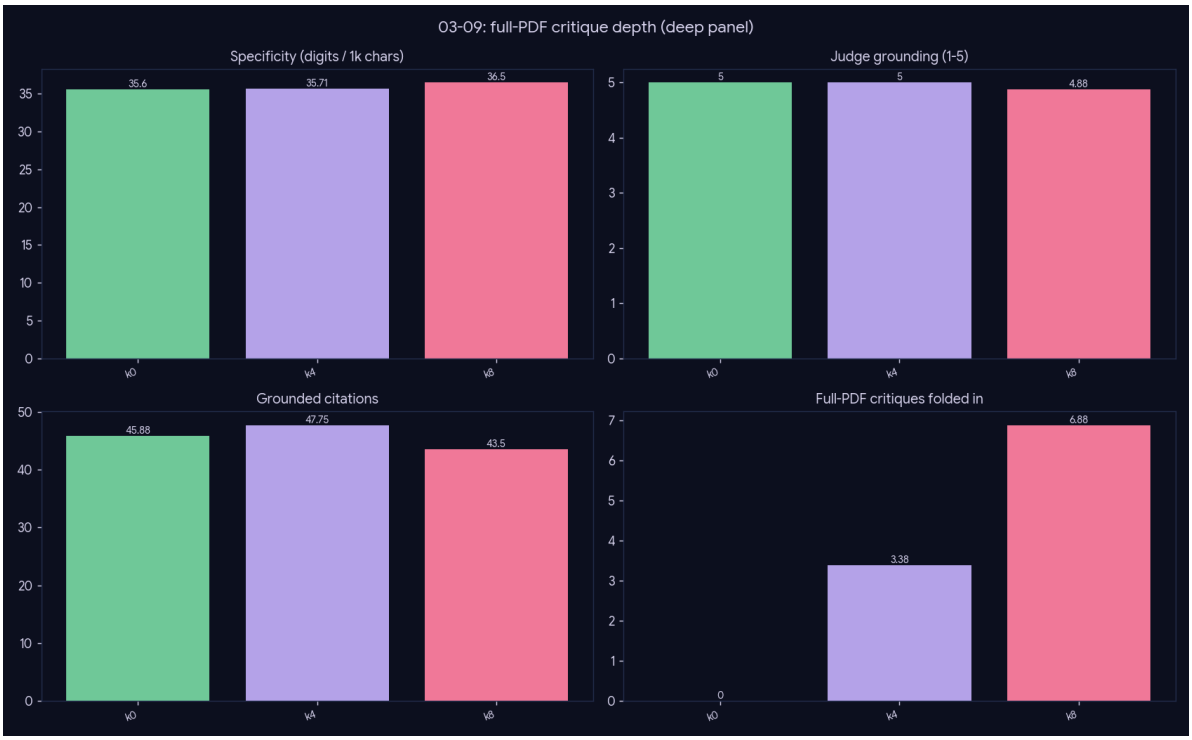
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What we did

Vary only `critique_k` $\in \{0, 4, 8\}$ — the number of per-paper full-PDF critiques the deep loop runs — holding everything else constant, over the 27-topic battery. Scored on the metric vector + 4-axis judge + faithfulness + an order-debiased **pairwise** of `k8` vs `k0`. The honest question: does reading more full PDFs measurably improve the review?

Results (mean over 27 topics)

critique_k	n critiques	words	num density	grounded	gold recall	j grounding	j coverage	j lineage	f
k0	0.0	3026.0	35.6	45.88	0.88	5.0	4.56	5.0	
k4	3.38	3216.12	35.71	47.75	0.92	5.0	4.62	4.81	
k8	6.88	3027.25	36.5	43.5	0.88	4.88	4.44	4.81	



Conclusion

A documented near-null. The first few critiques help a little — gold recall 0.88 → 0.92 from `k0` to `k4` — but `k8` **adds nothing over** `k4` on recall or the judge, and faithfulness *drops* (0.94 at

`k8` vs 0.95 at `k0`): more critiqued passages give the composer more chances to mis-attribute. The pairwise `k8` vs `k0` is a modest 0.63 overall — below the deep loop's 0.78 — and is not backed by any objective-metric gain. **Critique depth is not a quality lever beyond a handful**: keep `critique_k` small (≈ 4); paying for $k = 8$ buys cost (full-PDF reads) and a faithfulness risk, not quality. **Caveat:** this is the kind of honest null worth recording — the deep loop's win (`03-08`) comes from the *retrieve-and-expand* rounds, not from critiquing more PDFs per round.